Eschikon Plant Stress Phenotyping Dataset

dataset overview

Overview

We present a dataset with spatio-temporal-spectral data pertaining to sugarbeet crop growth under no, drought, fertiliser and weed stress conditions over two months. The dataset contains biweekly color images, infra-red stereo image pairs and multispectral camera images along with applied treatment parameters and temperature and humidity of the surroundings. If you find this dataset useful, please cite the following work

> A Spatio Temporal Spectral Dataset for Plant Stress Phenotyping by Khanna, Raghav and Schmidt, Lukas and Nieto, Juan and Siegwart, Roland and Liebisch, Frank

Treatments at a Glance

Description	# of Boxes	Soil Type	Water input	Nitrogen Input	Weed Pressure	Box Numbers
Control	3	2	Sufficient	High	None	1,2,3
Low N	3	1	Sufficient	Low	None	22,23,24
Med N	3	2	Sufficient	Medium	None	4,5,6
Med weeds	3	2	Sufficient	High	Medium	7,8,9
High weeds	3	2	Sufficient	High	High	10,11,12
Dry	3	2	Low	Medium	None	13,14,15
Weed Only Dicot	1	2	Sufficient	High	High	16
Weed Only Monocot	1	2	Sufficient	High	High	17
Weed Only Mixed	1	2	Sufficient	High	High	18
Low N - Med Weed	3	1	Sufficient	Low	Medium	25,26,27
Drying - Med N - High Weed	3	2	Low	Medium	High	19,20,21
Drying - Low N	3	1	Low	Low	None	28,29,30
CALTag and Color Checker	1	1	NA	NA	NA	31

Download

The entire dataset may be found here [http://robotics.ethz.ch/~asl-datasets/2018_plant_stress_phenotyping_dataset/].

Software providing common useful pre and post processing routines can be found in the accompanying github repository [https://github.com/ethz-asl/plant_stress_phenotyping].

The dataset consists of the following data. The folder structure is as follows, folder and file names should be indicative of the data they contain.

The images.zip (2.5GB) [http://robotics.ethz.ch/~asl-datasets/2018_plant_stress_phenotyping_dataset/images.zip] archive contains the raw image data. Images are organised by their measurement date and further by the camera (rgb/infra-red/multispectral)used.

The pointclouds.zip (2.5GB) [http://robotics.ethz.ch/~asl-datasets/2018_plant_stress_phenotyping_dataset/pointclouds.zip] archive contains the processed spectral point cloud data in `.mat` format. Point Clouds are indexed chronologically. Please refer to the `MeasurementPointCloud` class in the accompanying software for a detailed description of the data structure.

Geometric calibration images (containing a checkerboard) can be found in the calibration.zip [http://robotics.ethz.ch/~asl-datasets/2018_plant_stress_phenotyping_dataset/calibration.zip] archive.

The temperature and humidity logs from the greenhouse can be found in the greenhouse-temperature-humidity-logs.csv [http://robotics.ethz.ch/~asl-

datasets/2018_plant_stress_phenotyping_dataset/greenhouse-temperature-humidity-logs.csv] file.

The biweekly measurements of box weights present in the box-weights-kg.csv [http://robotics.ethz.ch/~asl-datasets/2018_plant_stress_phenotyping_dataset/box-weights-kg.csv] file can be used to estimate the soil moisture content at that time.

SPAD measurements can be found in the spad-measurements.csv [http://robotics.ethz.ch/~asl-datasets/2018_plant_stress_phenotyping_dataset/spad-measurements.csv] file.

Weights in *grams* of both above and below ground biomass for each box which serve as yield indicators may be found in the yield-weight-grams.csv [http://robotics.ethz.ch/~asl-datasets/2018_plant_stress_phenotyping_dataset/yield-weight-grams.csv] file.